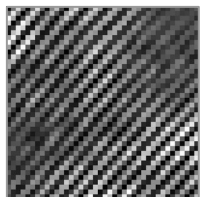
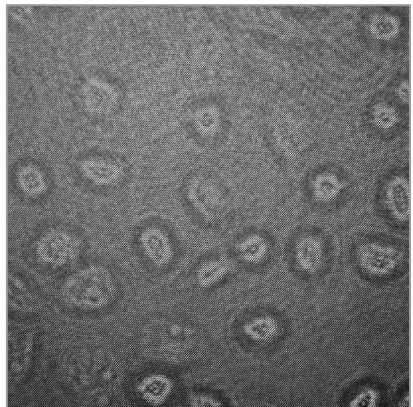
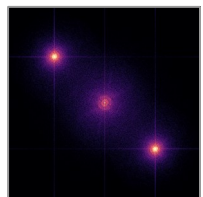


a Raw hologram input

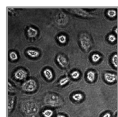
Off-axis hologram (one channel, 8-bit), cropped $1024^2 \rightarrow 900^2$ onto the phase grid (origin moved by $(-6, -2)$ px)



fringe detail



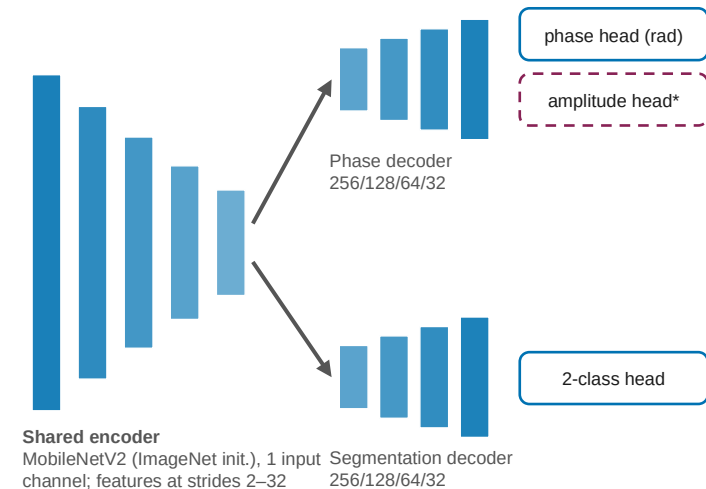
|FFT|: DC + sidebands



In-line (Gabor) hologram
same field; same architecture
trained separately (In-Line Neural
Configuration)

b End-to-end network

One shared encoder, two decoders; no classical reconstruction in the inference path



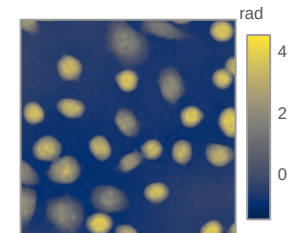
U-Net skip connections from every encoder stage; outputs at stride 2, bilinearly upsampled to 900^2

9.60 M parameters · 45.85 GMAC

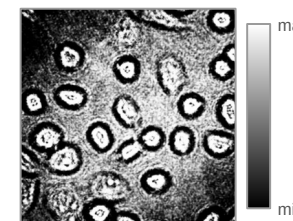
Compact: 256-ch projection, 3.36 M

* amplitude head only in +Amplitude and +Fwd configurations

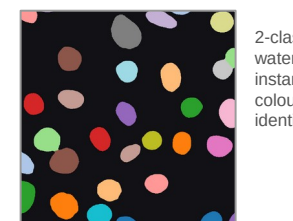
c Predicted fields



Quantitative phase (rad)



Transmitted amplitude*

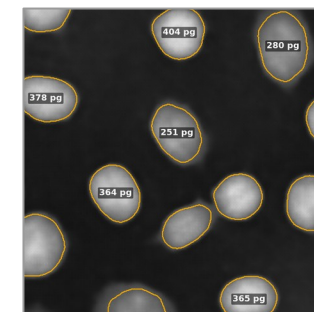


Cell segmentation (instances)

* +Amplitude / +Fwd only; agreement is with a reconstruction-derived amplitude, not a measured one

2-class map → watershed instances; colour = identity only

d Per-cell measurement



Cell k, predicted phase φ over its domain Ω_k :

$$A_k = N_k \, dx \, dy$$

$$C_k = \min(1, 4\pi \, N_k / P_k^2)$$

$$S_k = \sum \varphi \, dx \, dy$$

$$m_k = \lambda / (2\pi\alpha) \cdot S_k$$

$$\lambda = 0.666 \, \mu\text{m}$$

$$\alpha = 0.2 \, \text{mL/g}$$

$$dx = dy = 0.285 \, \mu\text{m}$$

Dry mass pg per cell (primary quantity)

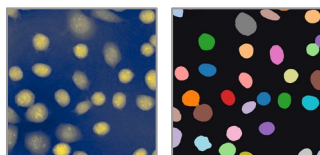
Projected area μm^2 per cell

Circularity shape descriptor, 0–1

Integrated phase S_k rad· μm^2 per cell (30–6000 μm^2 cells)

Counts: detection recall and precision reported alongside, because per-cell errors are computed only for detected and matched cells

e Supervision and training objective



labels are derived from the reference phase (no manual annotation)

Every configuration: $\mathcal{L}_{\text{phase}} = \text{L1} + 0.5 \cdot |\nabla| + 0.2 \cdot (1 - \text{SSIM})$ $\mathcal{L}_{\text{seg}} = \text{Dice} + \text{CE}$

Added one term at a time (ablations):

measurement-aware

+IPP (per-cell)

+IPP (image)

+Area

+BGA

output / forward model

+Amplitude

+Fwd (fixed / free z)

hologram forward model on $\hat{A} \cdot \exp(i\varphi)$

f Evaluation on the measurand

vs reference

per-cell MAPE over matched cells (IoU ≥ 0.5), with recall, coverage-adjusted MAPE and per-field Bland–Altman

vs Classical Pipeline

angular-spectrum reconstruction + the same segmentation labelling and measurement chain; off-axis and in-line

Model-free error floor

known boundary shifts propagated to area and mass; chain checked on analytic fields